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A. Packet Sniffing Using Raw Socket in Python

Aim: To implement a packet sniffer using raw sockets in Python that captures live network traffic and parses Ethernet, IP, and transport-layer (TCP/UDP/ICMP) headers.

Steps:

1. Set up the Lab Environment
2. Understand the Packet Structure You'll parse
3. Create the Raw Socket
4. Parse the Ethernet Header
5. Parse the IP header.
6. Parse TCP/UDP/ICMP headers
7. Write the Main Sniffing Loop.
8. Run the Sniffer
9. Extend the Lab
10. Lab Report - What to Document

Result:

Thus the Packet Sniffing using Raw Socket is implemented and executed successfully.

ne.py > ...

```python
import socket
import struct
sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
host = socket.gethostbyname(socket.gethostname())
sniffer.bind((host, 0))
sniffer.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
print("Packet Sniffer Started...\n")
try:
    while True:
        raw_data, addr = sniffer.recvfrom(65535)
        ip_header = raw_data[0:20]
        iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
        version = iph[0] >> 4
        ihl = iph[0] & 0xF
        ttl = iph[5]
        protocol = iph[6]
        source = socket.inet_ntoa(iph[8])
        destination = socket.inet_ntoa(iph[9])
        print("----------------------------------------")
        print("Version :", version)
        print("Header Length:", ihl * 4)
        print("TTL :", ttl)
        print("Protocol :", protocol)
        print("Source IP :", source)
        print("Destination :", destination)
        print("Packet Length:", len(raw_data))
except KeyboardInterrupt:
    sniffer.ioctl(socket.SIO_RCVALL, socket.RCVALL_OFF)
    print("\nStopped")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Packet Sniffer Started...

----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.168.56.1
Destination : 224.0.0.251
Packet Length: 56
----------------------------------------
Version : 4
Header Length: 20
TTL : 1
Protocol : 17
Source IP : 192.168.56.1
Destination : 224.0.0.251
```